Isolating Emotion from Motion: Classifying Affective States from Wearable Physiological Signals

Suhil Balamurugan
Diamond Bar High School
CalCyAIVR Research Internship Program
Diamond Bar, California

Dr. Husain
Research Mentor
CalCyAIVR Research Internship Program
Diamond Bar, California

***Abstract*—Wearable devices increasingly use physiological signals to infer emotional state, offering a path toward accessible stress and mental-health monitoring. However, the optical sensors these devices rely on are highly susceptible to motion artifacts, raising the concern that a classifier could exploit incidental body movement as a shortcut rather than genuine physiology. This work investigates whether machine learning models can classify affective state from wearable physiological signals, and whether such models depend on motion to do so. Using the WESAD dataset (15 subjects, three affective states: baseline, stress, and amusement), we extract 14 statistical features from 60-second windows of chest-worn ECG, electrodermal activity (EDA), respiration, temperature, and accelerometer signals, and evaluate Logistic Regression, Random Forest, and Support Vector Machine (SVM) classifiers under subject-independent, Leave-One-Subject-Out (LOSO) cross-validation. Logistic Regression achieves the best class-balanced performance (macro-F1 = 0.623), reliably separating stress but struggling with amusement, which physiologically resembles rest. A feature-ablation experiment shows that removing motion features does not reduce performance for any model, while Random Forest's own feature-importance ranking places a motion feature second overall – a reconciled result showing that motion is used when available but not necessary, because it is highly redundant with temperature variability ($r = 0.94$).**

***Index Terms*—affective computing, wearable sensors, physiological signals, emotion recognition, motion artifacts, machine learning, subject-independent validation**

I. Introduction

Wearable devices are increasingly used to monitor physiological state continuously and unobtrusively, creating an opportunity to detect stress and other emotional states outside of controlled clinical settings. Detecting affect from signals such as cardiac activity and skin conductance could make mental-health and stress monitoring accessible to anyone wearing a consumer smartwatch or fitness band, rather than requiring specialized laboratory equipment. This is significant because chronic, undetected stress is linked to a wide range of negative health outcomes, and early, passive detection could allow for earlier intervention.

A substantial body of research has shown that physiological signals – particularly electrocardiography (ECG), photoplethysmography (PPG), and electrodermal activity (EDA) – carry information about emotional and stress state, and that machine learning classifiers, most commonly support vector machines, can recover this information with reported accuracies frequently exceeding 90% [2]. The Wearable Stress and Affect Detection (WESAD) dataset, used in this work, was introduced with baseline classifiers achieving roughly 80% accuracy on a three-class affect problem and roughly 93% on binary stress detection [1]. However, nearly all of this reported performance comes from data collected while subjects were seated or otherwise still. The optical PPG sensors used in most consumer wearables are known to be highly sensitive to motion artifacts, with sensor-accuracy studies attributing a large share of measurement error to movement during recording [3], [4]. This creates an underexplored risk: an emotion classifier trained on such data could partly, or largely, be keying on incidental body motion – for example, more movement during an emotionally activating condition – rather than on the underlying physiological response, which would make the classifier brittle in any real-world setting where motion cannot be controlled.

This raises the central question motivating this work: can physiological signals classify emotional state under realistic, subject-independent conditions, and does body motion meaningfully contribute to that prediction, or does it act as a confound? Answering the second half of this question requires more than reporting a single accuracy number – it requires directly manipulating the presence of motion information and observing what changes.

In this work, we investigate whether machine learning models can classify three affective states – baseline, stress,

and amusement – from chest-worn physiological signals in the WESAD dataset, and we directly test whether the resulting classifiers depend on motion. We extract a set of statistical features from windowed signals, evaluate three classifiers (Logistic Regression, Random Forest, and SVM) under strict subject-independent cross-validation, and quantify the role of motion through both a feature-ablation experiment and model-derived feature importance. We find that classification performance does not depend on motion features, even though one model uses a motion feature heavily when it is available, and we reconcile this apparent tension through a redundancy in the underlying signals.

II. Related Work

Physiological emotion and stress recognition has been studied extensively using cardiac and electrodermal signals. Ismail et al. [2] systematically reviewed 92 studies using cardio-based signals for emotion recognition and found that ECG and PPG were the dominant modalities, that support vector machines were the most frequently used classifier, and that reported accuracies were frequently above 90%. Schmidt et al. [1] introduced the WESAD dataset used in this work, collecting chest and wrist physiological signals from 15 subjects under baseline, stress, and amusement conditions, and reported approximately 80% three-class accuracy and 93% binary stress-detection accuracy using hand-crafted statistical features with a random forest classifier.

A separate line of work has examined the reliability of the sensors themselves. Bent et al. [3] found motion to be a leading cause of error in wearable optical heart-rate sensors, and Elgendi [4] similarly analyzed how movement affects PPG-derived features. Other wearable stress-detection studies, such as Gjoreski et al. [7] and the review by Taskasaplidis et al. [6], likewise report that most evaluation is performed under low-motion, context-controlled conditions. Complementary literature establishes what these features measure: heart-rate variability (HRV) is a well-established marker of autonomic activity linked to stress [8], [9], Random Forest feature importance is a standard tool for interpreting nonlinear models [10], and deep-learning approaches such as DeepPPG [11] illustrate that model architectures are typically developed and tuned on similarly low-motion recordings.

Two patterns emerge that motivate the present study. First, almost all reported performance figures – including WESAD's own baseline – come from settings where movement is minimal, so it is largely untested whether these classifiers rely on motion-correlated artifacts rather than genuine physiology. Second, feature-importance rankings are rarely followed by an ablation test showing whether a highly-ranked feature is actually necessary, as opposed to merely associated with the outcome. This work differs from prior work by testing both angles at once: it evaluates a subject-independent classifier and quantifies motion's contribution with both an ablation experiment and a model-importance analysis, rather than relying on either method alone.

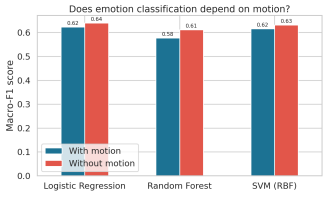


Fig. 1. Bar Chart

III. Dataset and Exploratory Data Analysis

A. Dataset

We use the WESAD (Wearable Stress and Affect Detection) dataset [1], collected from 15 subjects wearing a RespiBAN chest device (700 Hz: ECG, EDA, EMG, respiration, and temperature) and an Empatica E4 wrist device, while three affective conditions were induced: a neutral baseline, stress (induced via public speaking and mental arithmetic tasks), and amusement (induced via funny video clips). We use the chest-worn signals in this work.

Because the raw 700 Hz recordings are far too dense to use as direct classifier inputs, we segment each subject's chest signals into non-overlapping 60-second windows and compute summary statistics (mean, standard deviation, and related measures) for each recorded channel (heart rate, EDA, respiration, temperature, and three-axis accelerometry), yielding 16 raw features per window. Two EDA-derived features were found to be perfectly correlated ($r = 1.00$) with existing features and were dropped, leaving 14 features. This process produced 520 labeled windows: 283 baseline, 155 stress, and 82 amusement – a class-imbalance ratio of approximately 3.4:1 between the largest and smallest class, which we account for during modeling (Section IV). All inputs are continuous numerical values; the target is a categorical three-class label. The dataset contains no missing values.

B. Exploratory Findings

Fig. 6 shows the distribution of representative features across the three affective conditions. Stress is clearly separable: heart rate, respiration variability, and electrodermal activity are all visibly elevated relative to the other two conditions. Amusement, by contrast, overlaps substantially with baseline across nearly every feature, foreshadowing that amusement would be the most difficult class to classify – physiologically, watching a brief amusing video does not appear to produce as strong an autonomic response as a stressful task does.

To understand which features separate the three classes best in a purely statistical sense, we ranked all 16 raw features by their ANOVA F-value (Fig. 7). Heart-rate-derived features dominate the top of the ranking, followed by respiration variability, while the accelerometer-derived motion feature (ACC_std) ranks 7th of 16 – informative, but solidly mid-table rather than dominant. This is an important early observation: motion is measurably associated with class separation, which is precisely the condition under which a classifier could end

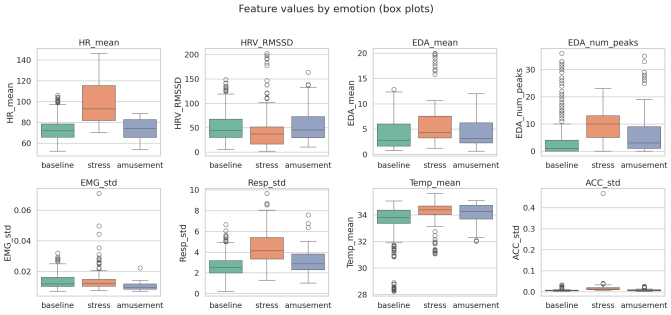


Fig. 6. Distributions of representative physiological features by affective condition. Stress is elevated across heart rate, respiration variability, and EDA; amusement overlaps heavily with baseline.

up depending on it. This observation motivated the motion-ablation experiment described in Sections IV and V, which tests that dependence directly rather than inferring it from a ranking alone.

IV. Methodology

A. Data Preprocessing

No missing-value imputation was required, as the windowed feature set contained no missing entries. Two EDA-derived features that were perfectly redundant with other retained features ($r = 1.00$) were removed prior to modeling, leaving 14 features. All features were standardized to zero mean and unit variance prior to training, since the classifiers used here – particularly Logistic Regression and SVM – are sensitive to differences in feature scale. No categorical encoding was required, as all input features are continuous; the target label is categorical and was left unencoded for classifiers that support multi-class labels directly.

B. Feature Selection and Engineering

The 14 retained features summarize five signal modalities per 60-second window: cardiac (heart-rate mean and variability), electrodermal activity, respiration, peripheral temperature, and three-axis accelerometry (motion). No additional engineered features (e.g., frequency-domain HRV measures) were computed in this iteration; feature relevance was instead assessed using univariate ANOVA F-values (Section III-B) and, for one model, impurity-based feature importance (Section V-C). The accelerometer-derived features (ACC_mean, ACC_std) were deliberately retained in the primary feature set so that their contribution could be measured and later ablated, rather than removed a priori – removing a feature of interest before testing it would defeat the purpose of this study.

C. Machine Learning Models

We evaluated three classifiers, chosen to span a range of model complexity and to match common practice in the emotion-recognition literature: (1) *Logistic Regression*, a linear model that serves as an interpretable, low-variance baseline; (2) *Random Forest* (300 trees), a nonlinear ensemble method capable of capturing feature interactions and, as a byproduct of training, providing an impurity-based feature-importance ranking; and (3) *Support Vector Machine* with a radial basis function (RBF) kernel, the most commonly reported classifier in prior physiological emotion-recognition work [2] and one that tends to perform well with a modest number of samples. All three models were trained with class weights inversely proportional to class frequency to counteract the roughly 3.4:1 class imbalance described in Section III-A; without this adjustment, models tend to default toward predicting the majority (baseline) class.

D. Experimental Setup

Rather than a single train/test split, we evaluated all models using Leave-One-Subject-Out (LOSO) cross-validation: for each of the 15 subjects, a model is trained on the windows from the remaining 14 subjects and tested on the held-out subject's windows, repeated so that every subject serves as the test set exactly once. LOSO was chosen deliberately over a random split because physiological baselines vary substantially between individuals (e.g., resting heart rate); a random split would allow a model to partly memorize subject-specific patterns rather than learning generalizable affect signatures, which would overstate real-world performance.

All experiments were implemented in Python using scikit-learn [5] for modeling, with NumPy, pandas, SciPy, and NeuroKit2 used for signal processing and feature extraction, and Matplotlib/seaborn for visualization, in a Google Colab environment. Given the modest dataset size, hyperparameters were kept at standard defaults (with the Random Forest tree count set to 300) rather than tuned per fold, to avoid overfitting the already-limited amount of subject-level data. We report accuracy and macro-averaged F1-score (the unweighted mean of per-class F1) for every model; macro-F1 is treated as the primary metric because, under class imbalance, accuracy can be inflated by a model that performs well only on the majority class (Section V). Fig. 8 summarizes the full pipeline from raw signal to evaluation.

V. Results and Discussion

A. Model Comparison

Table I reports LOSO accuracy and macro-F1 for all three classifiers. Random Forest achieves the highest raw accuracy (0.731) but the lowest macro-F1 (0.577); this discrepancy is not a contradiction but a warning sign. Because baseline is the largest class, a classifier can achieve a deceptively high accuracy simply by favoring it. Inspecting Random Forest's per-class behavior confirms this: its amusement recall was only 0.07, meaning it correctly identified almost none of the amusement windows, while still scoring well on the far larger baseline and stress classes. This is exactly the scenario macro-F1 is designed to expose, and it is why macro-F1, not accuracy, is treated as the primary metric throughout this work.

By macro-F1, Logistic Regression performs best (0.623), narrowly ahead of SVM (0.616) and clearly ahead of Random Forest (0.577). This is a meaningful result in its own right: the

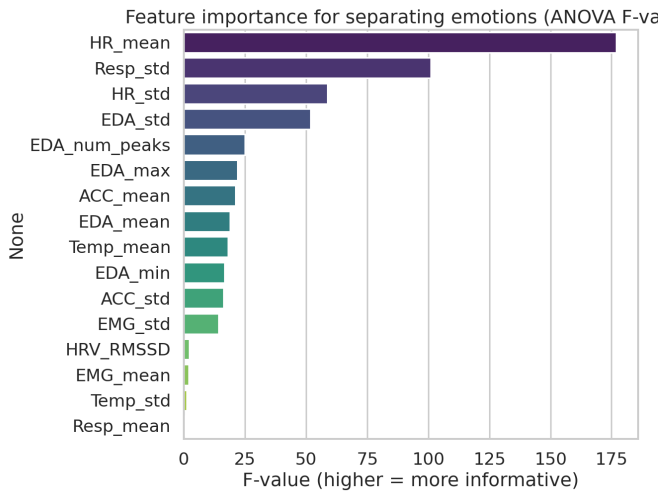


Fig. 7. Features ranked by ANOVA F-value. Heart-rate features dominate; the motion feature (ACC_std) ranks 7th of 16.

TABLE I
MODEL PERFORMANCE (LEAVE-ONE-SUBJECT-OUT)

Model	Accuracy	Macro-F1
Logistic Regression	0.648	**0.623**
Random Forest	**0.731**	0.577
SVM (RBF)	0.629	0.616

simplest of the three models is also the fairest across classes, suggesting that once class imbalance is properly weighted, the relationship between these physiological features and affective state is close to linearly separable, at least for the more distinguishable classes.

B. Per-Class Performance and the Confusion Matrix

Table II shows that stress is detected reliably (F1 = 0.74, recall = 0.79), consistent with the clear separation observed during EDA (Section III-B). Amusement is the hardest class by a wide margin (F1 = 0.46, precision = 0.37): the confusion matrix (Fig. 9) shows 74 baseline windows misclassified as amusement, mirroring the near-total overlap between these two classes observed in the box-plot distributions (Fig. 6). This is a physiologically sensible failure mode rather than a modeling defect – a brief amusing video clip does not appear to produce an autonomic response strong enough to reliably distinguish it from quiet rest using these features.

Compared to the $\sim$80% three-class accuracy reported in the original WESAD study [1], our results are lower on raw accuracy but were obtained under a considerably stricter, fully subject-independent evaluation; we consider macro-F1 = 0.623 an honest, conservative estimate of real-world, unseen-user performance rather than a shortfall relative to prior work.

C. Testing the Motion Confound

The central question of this work is whether the classifiers described above depend on motion. We tested this in two

TABLE II
PER-CLASS PERFORMANCE (LOGISTIC REGRESSION)

Class	Precision	Recall	F1
Baseline	0.79	0.58	0.67
Stress	0.71	0.79	0.74
Amusement	0.37	0.61	0.46

complementary ways.

First, we ran a feature-ablation experiment, retraining every model with the two accelerometer-derived motion features removed and comparing macro-F1 to the full-feature result. As shown in Fig. 10, removing motion did not reduce performance for any model; macro-F1 was unchanged or marginally higher in every case (Logistic Regression: 0.623 to 0.640; Random Forest: 0.577 to 0.612; SVM: 0.616 to 0.632). If the classifiers were relying on motion as a shortcut, this ablation should have produced a clear performance drop; instead, performance was, if anything, slightly better without it.

Second, we examined Random Forest's impurity-based feature importance on the full feature set (Fig. **??**). Here the accelerometer feature ACC_std ranks as the second most important feature overall – a result that, taken alone, appears to directly contradict the ablation finding. We resolve this apparent contradiction by noting that ACC_std is highly correlated ($r = 0.94$) with peripheral temperature variability (Temp_std), one of the retained physiological features. Random Forest is free to split on either of two highly correlated features and will often use whichever is available first, inflating that feature's importance score without making it uniquely necessary; once ACC_std is removed, Temp_std and the remaining physiological features carry equivalent information, so no signal is actually lost.

We summarize this as: motion is used when available, but not needed. This is consistent with, and helps explain, why the ANOVA ranking in Section III-B placed motion only 7th of 16 rather than at the very top – motion carries real but redundant information, and the model's reliance on it when present is an artifact of feature correlation rather than evidence that motion drives the underlying classification. The broader methodological lesson is that a feature's importance score and its necessity are distinct properties, and that a definitive test of confounding requires an ablation experiment, not an importance ranking alone.

VI. LIMITATIONS AND FUTURE WORK

This study has several limitations. First, WESAD's activity conditions are almost entirely seated, so the finding that motion does not confound classification may not generalize to high-motion settings such as walking or exercising; a natural extension is to repeat this ablation on a high-motion dataset such as PPG-DaLiA. Second, the dataset is small – 520 windows from only 15 subjects, with just 82 amusement windows – which limits statistical power for the already-difficult amusement class and makes individual LOSO folds noisy. Third, feature engineering was limited to time-domain

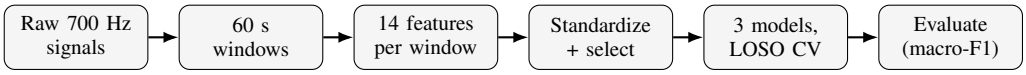


Fig. 8. Analysis pipeline, from raw chest-sensor signal to subject-independent evaluation.

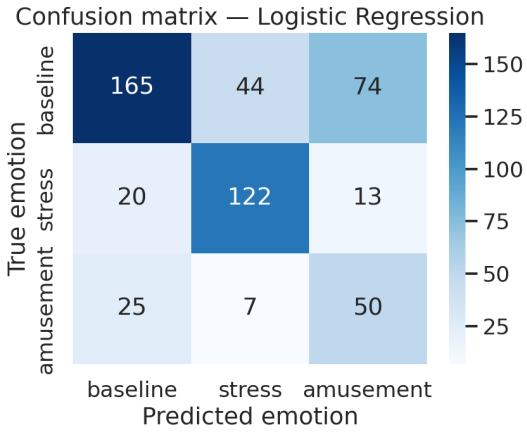


Fig. 9. Confusion matrix, Logistic Regression (LOSO). Amusement is frequently confused with baseline.

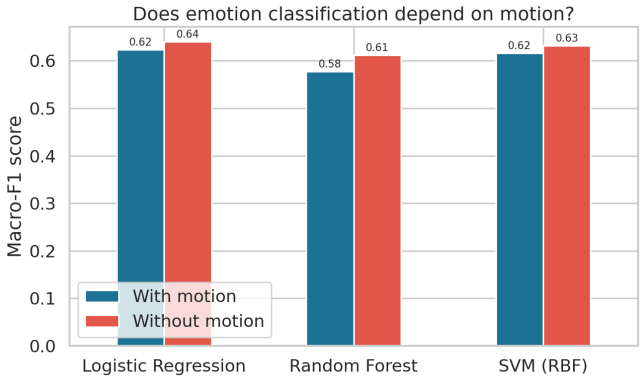


Fig. 10. Macro-F1 with vs. without motion features. Performance holds, or slightly improves, once motion is removed for all three models.

statistics over 60-second windows; frequency-domain and nonlinear HRV measures, which typically require longer windows, were not computed and may improve separability, especially for amusement. Fourth, all features were standardized using statistics computed once across the full dataset prior to cross-validation, rather than being refit within each LOSO fold; since the scaling parameters are simple means and standard deviations rather than fitted predictive parameters, we expect a negligible effect on the reported metrics, but refitting per fold is the methodologically stricter approach. Finally, results come from a single dataset with no external validation set, so absolute performance should be read as specific to WESAD's population and induction protocol rather than as a general-purpose benchmark.

Future work should prioritize re-running the motion-ablation test on a high-motion dataset, incorporating frequency-domain and nonlinear HRV features, exploring gradient-boosted and neural-network models for the amusement class, and moving toward an on-device, real-time implementation for practical stress monitoring.

VII. Conclusion

This work investigated whether physiological signals from a chest-worn wearable can classify affective state under realistic, subject-independent conditions, and whether such classification depends on body motion. Using the WESAD dataset and Leave-One-Subject-Out cross-validation, Logistic Regression achieved the best class-balanced performance (macro-F1 = 0.623), reliably detecting stress while struggling with amusement, which closely resembles physiological rest. A feature-ablation experiment showed that removing motion features does not reduce performance for any model, even though Random Forest's own feature-importance ranking places a motion feature second overall – a tension resolved by that feature's strong correlation with temperature variability. The overall conclusion is that motion is used when available but is not necessary: the classifiers evaluated here rely on genuine physiological signal rather than incidental movement, at least within WESAD's largely stationary recording conditions. This suggests that concerns about motion as a confounding shortcut, while reasonable to test, did not materialize in this setting.

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